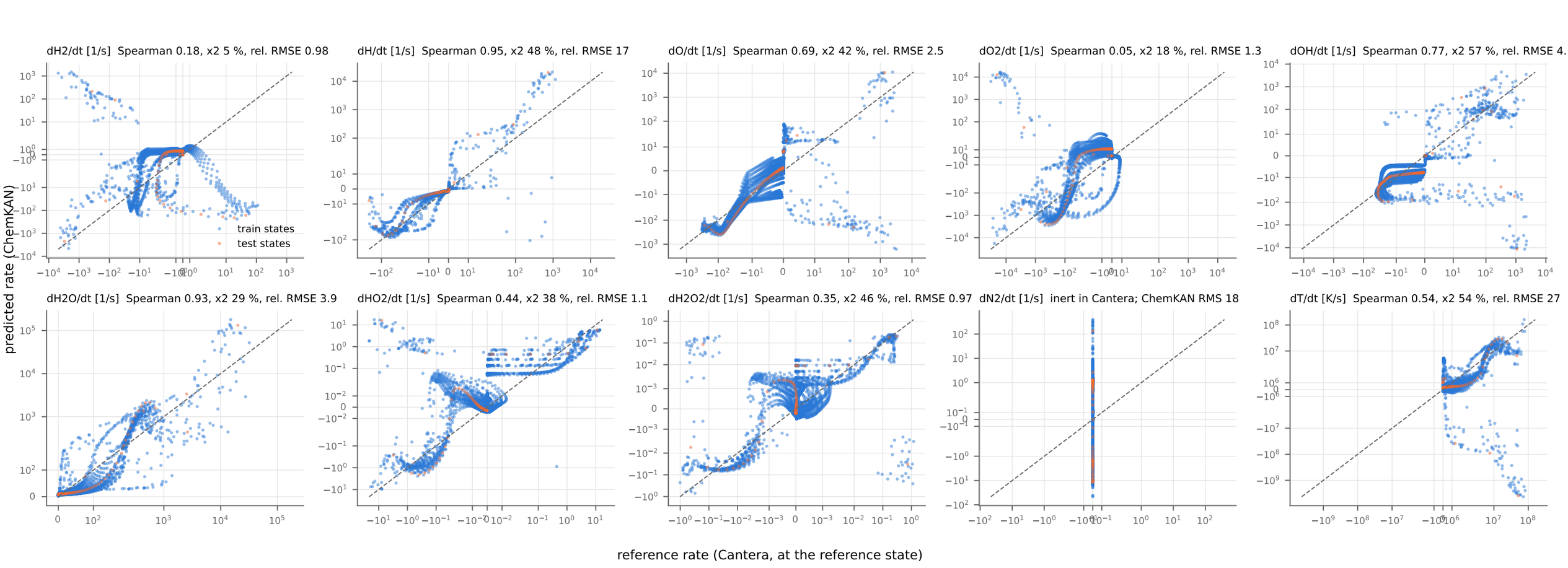

hydrogen: instantaneous rates at every state of the reference trajectories (symlog axes). Training states: rank correlation, share within a factor 2 (states with |rate| > 1 % of the species' max), relative RMSE
dH2/dt [1/s] Spearman 0.18, x2 5 %, rel. RMSE 0.98
dH/dt [1/s] Spearman 0.95, x2 48 %, rel. RMSE 17
dO/dt [1/s] Spearman 0.69, x2 42 %, rel. RMSE 2.5
dO2/dt [1/s] Spearman 0.05, x2 18 %, rel. RMSE 1.3
dOH/dt [1/s] Spearman 0.77, x2 57 %, rel. RMSE 4.
dH2O/dt [1/s] Spearman 0.93, x2 29 %, rel. RMSE 3.9
dHO2/dt [1/s] Spearman 0.44, x2 38 %, rel. RMSE 1.1
dH2O2/dt [1/s] Spearman 0.35, x2 46 %, rel. RMSE 0.97
dN2/dt [1/s] inert in Cantera; ChemKAN RMS 18
dT/dt [K/s] Spearman 0.54, x2 54 %, rel. RMSE 27
train states
test states
predicted rate (ChemKAN)
reference rate (Cantera, at the reference state)